BIOMEDICAL LIMITED

ANNUAL PRODUCT QUALITY REVIEW

(YEAR – 2025)

Protocol No. : **BML/QA/APQR/QA/002**

Protocol Date : **23/04/2024**

Total No. of Pages :

PRODUCT NAME : Ciprofloxacin Intravenous Injection 200mg/100ml

GENERIC NAME : Injection Ciprofloxacin (intravenous) BP

COMPOSITION : Each 100ml contains
Ciprofloxacin BP 200mg
Water for injection B.P q.s

PACK SIZE : 100ml in a printed PVC pouch

PACKAGING TYPES AVAILABLE : 100ml Polyvinylchloride PVC pouch
Corrugated cardboard boxes carton

MASTER FORMULAR NUMBER : BML/QA/MFR/017

SHELF LIFE: 3 Years

TIME PERIOD COVERED : JANUARY – DECEMBER, 2025

PRODUCT CODE : CP

BIOMEDICAL LIMITED

1, Ohimege Road, Industrial Estate, Gaa- Imam, Ilorin Kwara State Nigeria.

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Sr. No	Subject	Name	Designation	Sign/Date
1	Compiled by	Olawale Ogunleye	Quality Assurance Officer	
2	Reviewed by	Robiu Idowu M.	Production Pharmacist	
3	Approved by	Koroma Abigael A.	Quality Assurance Manager	
4	Authorized by	Agboola L. A	Factory Manager	

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1.0 SCOPE:

This covers internal annual products quality review, and all these parameters are monitored; changes in physical appearance of the product, chemical analysis (pH, Assay), microbiological analysis (Sterility and Pyrogen), and post market surveillance.

2.0 DESCRIPTION OF PRODUCT

Ciprofloxacin 200mg/100ml Intravenous Injection is a clear colorless sterile solution in a PVC pouch with double port for easy administration. Formulated for the treatment of infections due to gram-negative and gram-positive bacteria, the clear solution's manufacturing process is carefully carried out to ensure the appropriate pH of the solution and a label claim of 200mg/100ml throughout the shelf-life is achieved. The product batch detailing is heat-printed on the pouch to carry the batch number, manufacturing date and expiry date. Careful control of all other parameters as mentioned in the scope and under the section for the review of the quality control of the product in this piece is assured at all times. The product's NAFDAC registration number is A11-1190, with a packing size of 100mL.

3.0 REVIEWS

3.1 REVIEW OF PRODUCT BATCHES MANUFACTURED

A batch of Ciprofloxacin 200mg/100ml Intravenous Injection comprises of about 11,250 bags. Total No. of Batches produced within the period January 2025 to December 2025 is 8 batches. I.e. from Batch no.:CP25001 to CP25008; all the batches produced have shown reliability of the manufacturing and the testing procedures in delivering constant and reproducible good product quality.

3.2 REVIEW OF STARTING AND PRIMARY PACKAGING MATERIALS USED IN THE FPP

The starting materials for the manufacture of Ciprofloxacin 200mg/100ml Intravenous Injection are Ciprofloxacin powder (active ingredient), Sodium Chloride (active ingredient) and Water for injection (excipient). Ciprofloxacin and Sodium Chloride are sourced from approved vendors while water for injection is produced in house. The starting materials are each tested to comply with respective pharmacopeia and in-house specifications. The table in section 3.2.2, shows the different consignment of ciprofloxacin and Sodium Chloride received and used in the manufacture of Ciprofloxacin Intravenous Injection within the year under review. All the batches of the active material received met the specifications of pharmacopeia.

3.2.1 VENDORS

All the raw materials used in the production of these batches are sourced from the approved vendors only.

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3.2.2 RAW MATERIAL (API) MANUFACTURER/ SUPPLIER PERFORMANCE REVIEW

Name of API	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignment / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
Ciprofloxacin	Zheijang Guobang Pharmaceutical, China	Bdv Behrens, Germany	**1**	**1**	**Nil**	**Nil**
Sodium Chloride	K+S Minerals & Agriculture.	Fzco Dubai Silicon, Oasis.	**1**	**1**	**Nil**	**Nil**
	K+S Minerals & Agriculture.	K+S Minerals & Agriculture.	**1**	**1**	**Nil**	**Nil**
	Hebei Huachen Pharmaceutical Co. Ltd, China.	Hebei Huachen Pharmaceutical Co. Ltd, China.	**2**	**2**	**Nil**	**Nil**
	Salonen, Austria.	Muab Chemicals, Lagos.	**1**	**1**	**Nil**	**Nil**
	K+S Minerals & Agriculture.	Bdv Behrens, Germany	**1**	**1**	**Nil**	**Nil**

3.2.3 PRIMARY PACKAGING MATERIAL

The FPP is contained in poly vinyl chloride (PVC) self-collapsible soft bag (pouch) bought as a finished product from the manufacturer of the PVC pouch. Manufacturer COA of the PVC is reviewed and testing done in-house to confirm the quality specifications stated by the manufacturers against in-house/pharmacopoeial specifications set for the primary packaging material. The consignments received are as follows:

Name of Primary packaging Material	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignments / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
PVC pouches	Jiangsu Medwell Medical	Jiangsu Medwell Medical	13	13	Nil	Nil

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	Equipment Co. Ltd. China	Equipment Co. Ltd. China				

3.3 REVIEW AND STATISTICAL ANALYSIS OF QUALITY CONTROL AND IN-PROCESS CONTROL

3.3.1 SPECIFICATIONS

Appearance: Clear, colorless and particles free solution

Ciprofloxacin B.P.: 95 – 105%

Sodium Chloride B.P: 95 – 105%

pH: 3.90 – 4.50

3.3.2 IN-PROCESS RESULTS:

The summary of in-process results of all the batches manufactured was prepared and the table showing the results presented. The summary of in-process results for the chemical analysis and microbiological analysis of the batches manufactured is as shown below.

Chemical Analysis of In-process product Result

S/N	BATCH NO	Analysis			
		Ciprofloxacin Assay (95 – 105%)	**Sodium Chloride** Assay (95 – 105%)	**pH** (sample) (3.9 – 4.5)	**pH** (WFI) (5.0 – 7.0)
1	CP25001	99.91	98.2	4.38	6.4
2	CP25002	99.85	97.93	4.35	6.27
3	CP25003	98.32	97.93	4.4	6.19
4	CP25004	99.54	97.93	4.39	6.01
5	CP25005	98.31	97.31	4.33	6.07
6	CP25006	98.18	96.51	4.38	6.06
7	CP25007	99.81	97.78	4.17	6.03
8	CP25008	99.51	98.63	4.35	6.1

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Chemical/Microbial Analysis for Finished product Result

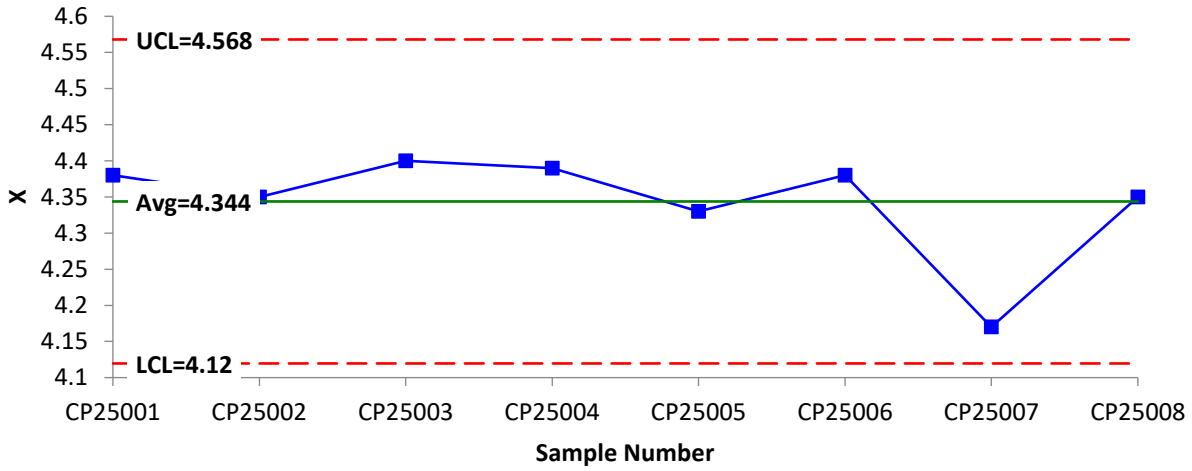
S/N	BATCH NO	**Ciprofloxacin** Assay (95 – 105%)	**Sodium Chloride** Assay (95 – 105%)	**pH** (3.9 – 4.5)	Sterility (MUST BE STERILE)	**Pyrogen** (1.15^{O}C)
1	CP25001	100.55	98.2	4.46	STERILE	<0.125
2	CP25002	99.04	99.19	4.03	STERILE	0.15
3	CP25003	98.24	96.68	4.37	STERILE	<0.125
4	CP25004	96.76	96.68	4.06	STERILE	<0.125
5	CP25005	99.48	97.93	4.08	STERILE	<0.125
6	CP25006	97.33	97.5	4.43	STERILE	0.4
7	CP25007	101.44	97.62	4.14	STERILE	<0.125
8	CP25008	100.71	98.48	4	STERILE	<0.125

3.3.3 TREND ANALYSIS

Trend analysis of Ciprofloxacin content, Sodium Chloride content, pH, Pyrogen and Final yield %, were carried out on all the above batches.

Annexure I: pH trend for inprocess analysis of the product

Specification (3.9 -4.5**)**



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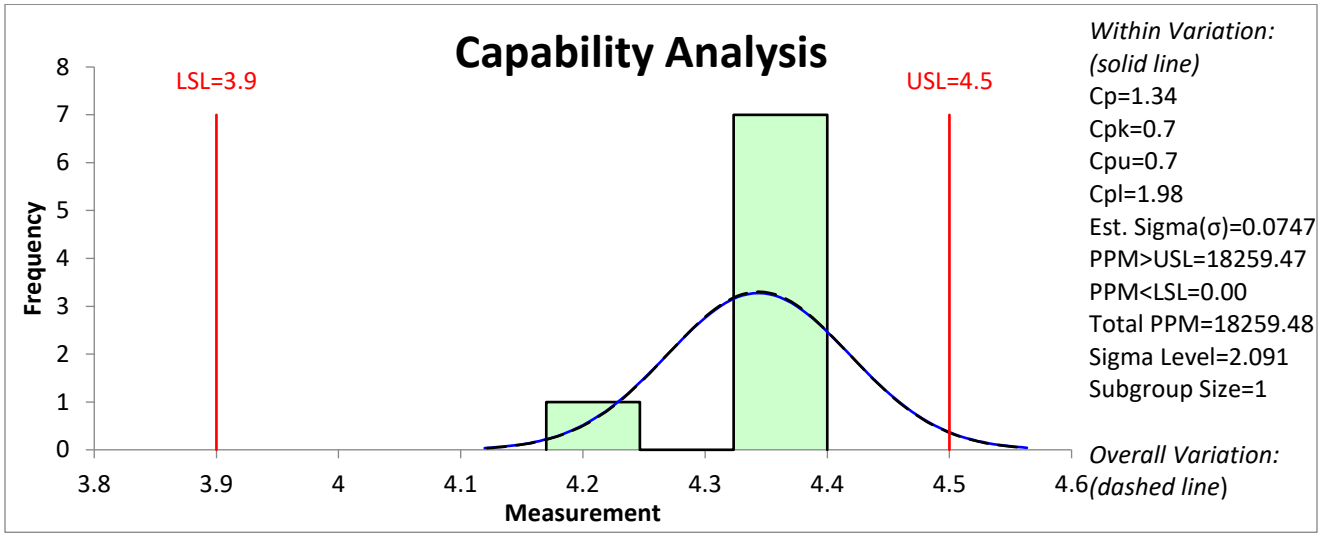
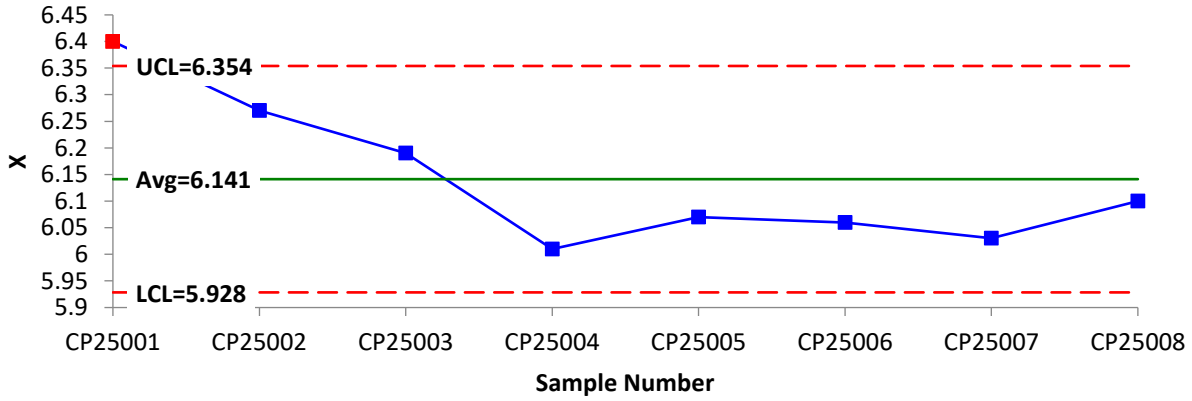
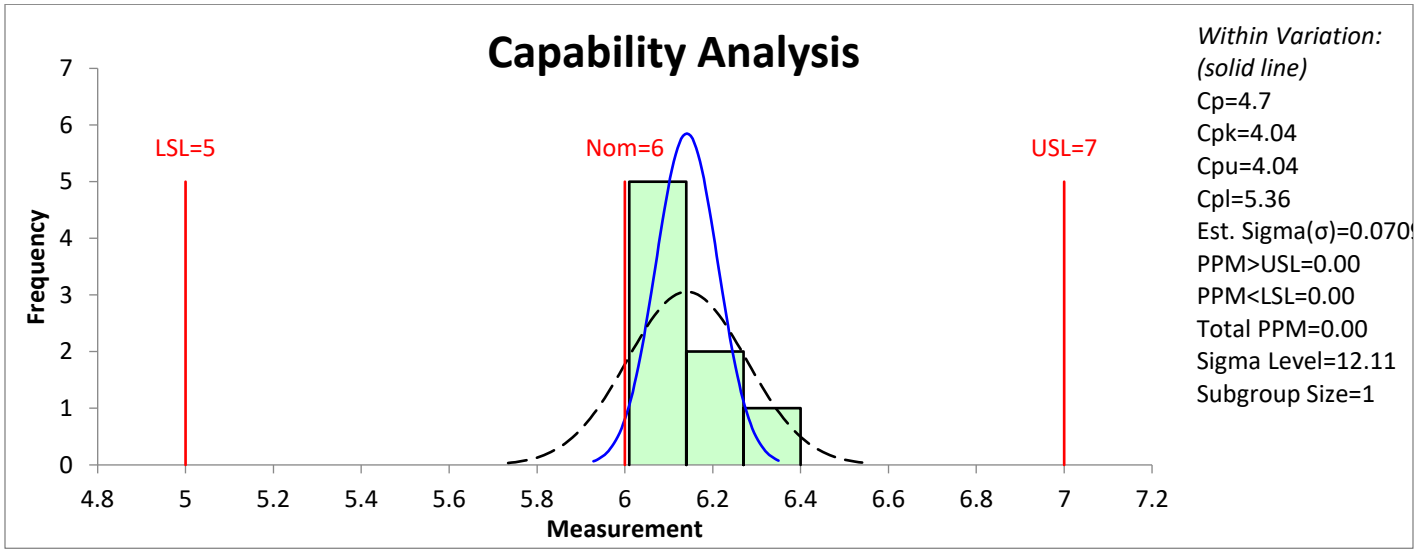


Mean	4.344
Standard deviation	0.074053552
Maximum	4.4
Minimum	4.17
Control limit: 4.12– 4.57	
Alert limit: 4.20 – 4.49	
Minimum-Maximum: 4.17 – 4.4	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

Annexure II: pH trend for Water for Injection (WFI)

Specification (5.0 -7.0)

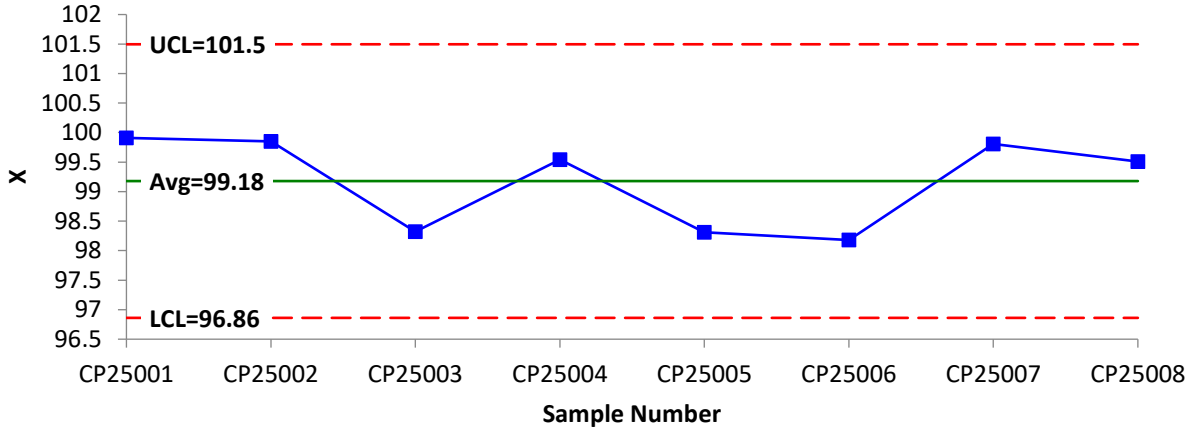


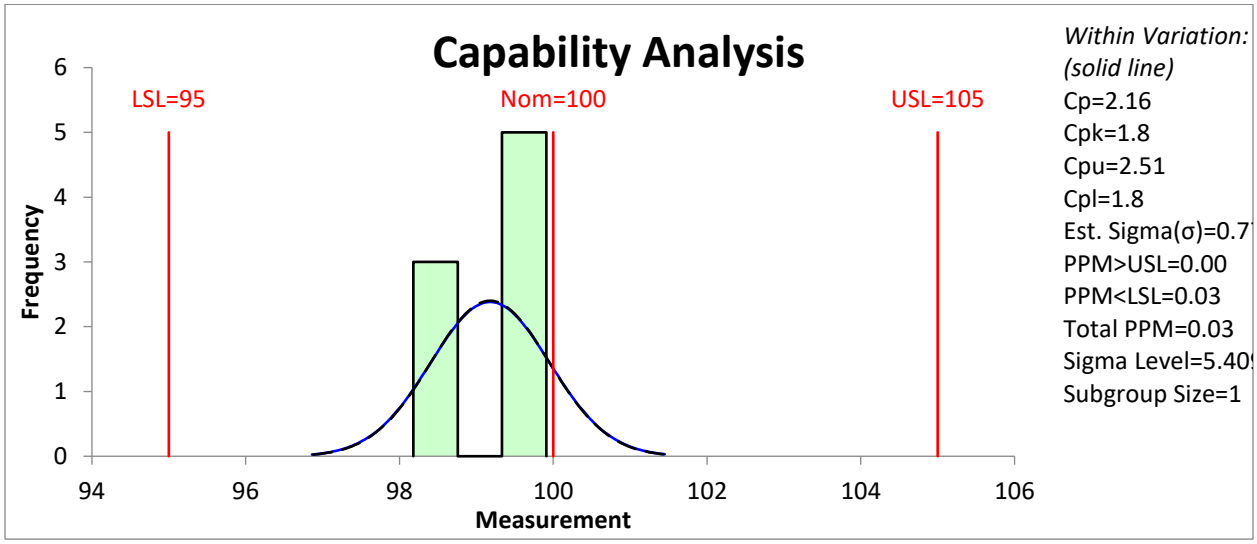


Mean	6.141
Standard deviation	0.135692667
Maximum	6.4
Minimum	6.01
Control limit: 5.928– 6.354	
Alert limit: 5.870 – 6.412	
Minimum-Maximum: 6.01– 6.4	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

Annexure III: Assay trend of in-process analysis of Ciprofloxacin in Ciprofloxacin Injection

Specification of Ciprofloxacin (95 – 105%**)**





Mean	99.18
Standard deviation	0.766596141
Maximum	99.91
Minimum	98.18
Control limit: 96.86– 101.5	
Alert limit: 97.65– 100.71	
'Minimum-Maximum: 98.18– 99.91	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

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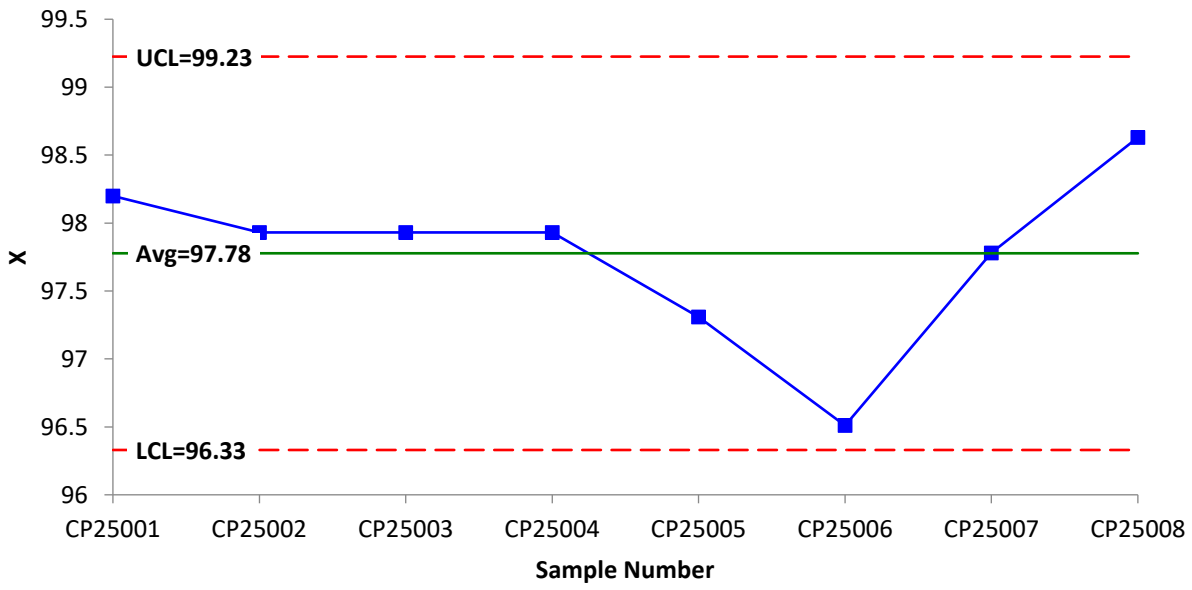
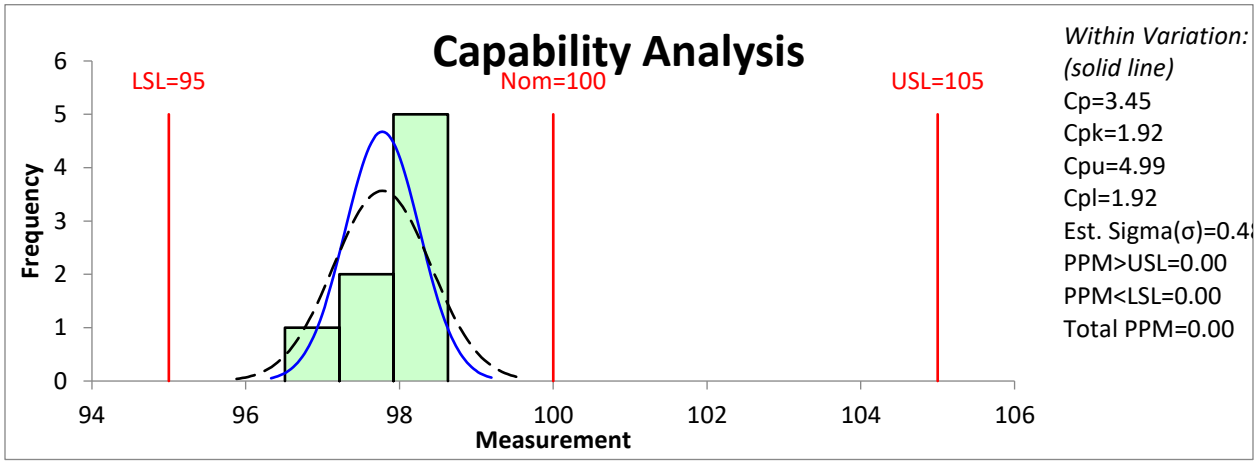
Annexure IV: Assay trend of in-process analysis of Sodium Chloride in Ciprofloxacin Injection

Specification of Sodium Chloride (95 – 105%**)**





Mean	97.78
Standard deviation	0.632472473
Maximum	98.63
Minimum	96.51
Control limit: 96.33– 99.23	
Alert limit: 96.52 – 99.04	
'Minimum-Maximum: 96.51– 98.63	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

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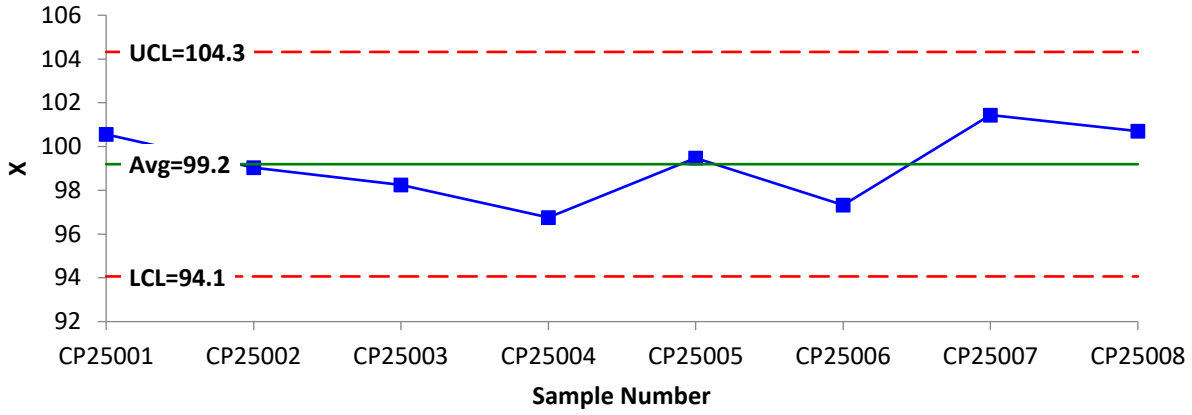
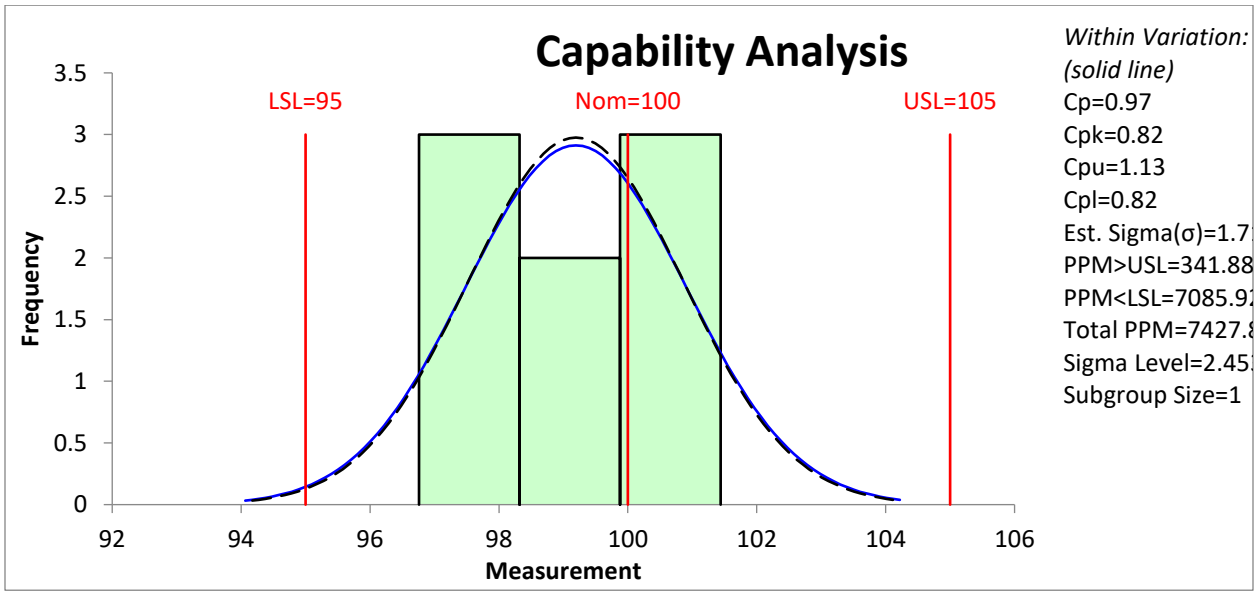
Annexure V: Assay trend of finished product analysis of Ciprofloxacin in Ciprofloxacin Injection

Specification of Ciprofloxacin (95 – 105%**)**





Mean	99.2
Standard deviation	1.673148764
Maximum	101.44
Minimum	96.76
Control limit: 94.1 – 104.3	
Alert limit: 95.85 – 102.55	
Minimum-Maximum: 96.76–101.44	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

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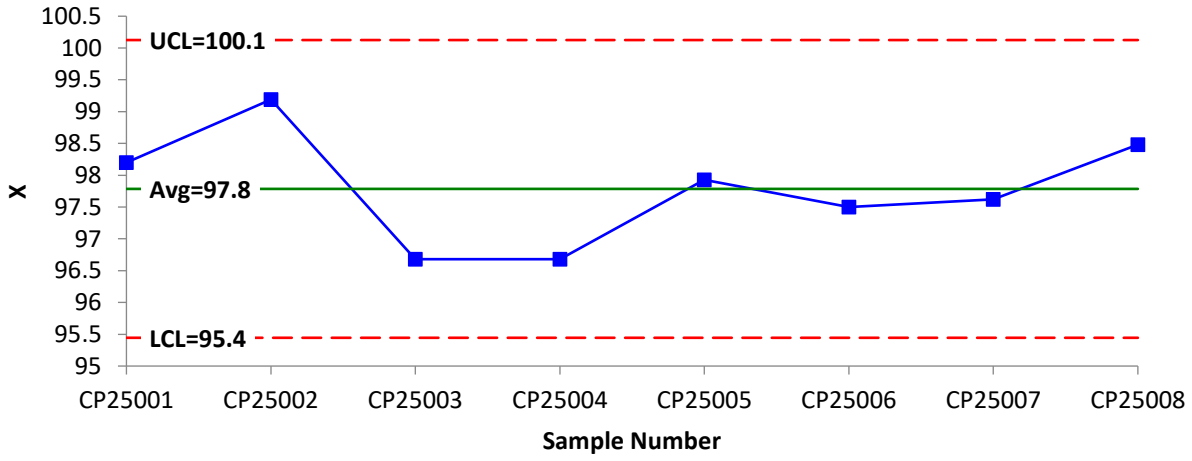
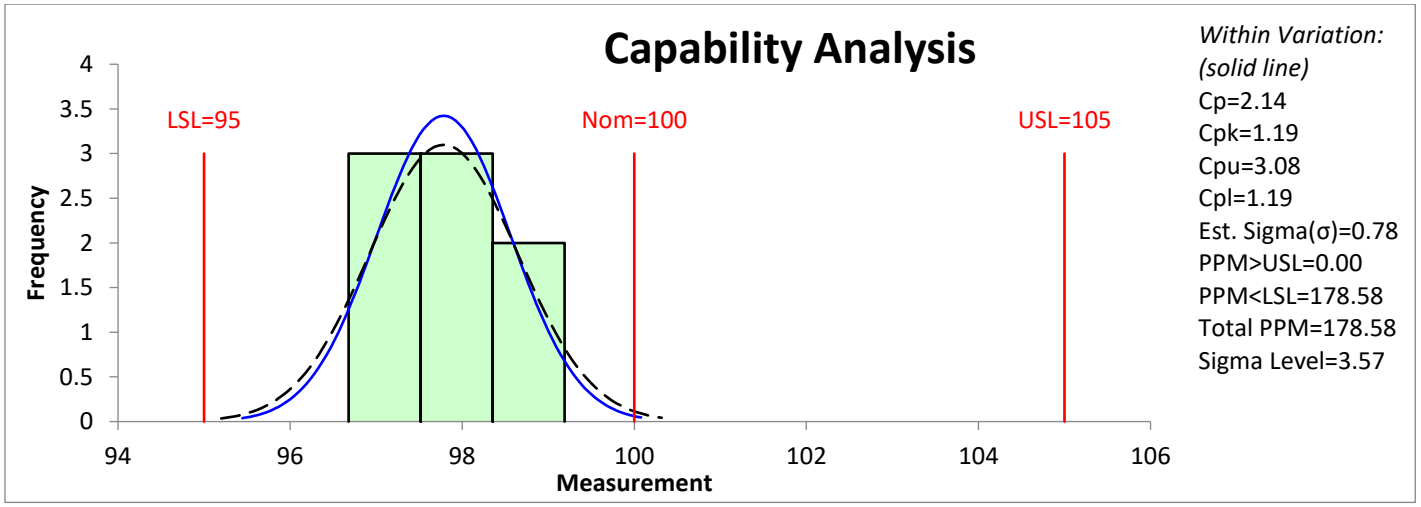
Annexure VI: Assay trend of finished product analysis of Sodium Chloride in Ciprofloxacin Injection

Specification of Sodium Chloride (95 – 105%)





Mean	97.8
Standard deviation	0.861957904
Maximum	99.19
Minimum	96.68
Control limit: 95.4– 100.1	
Alert limit: 96.08 – 99.52	
Minimum-Maximum: 96.68–99.19	
Alert limit within the specification limit, control limit existing within the specification limit.	

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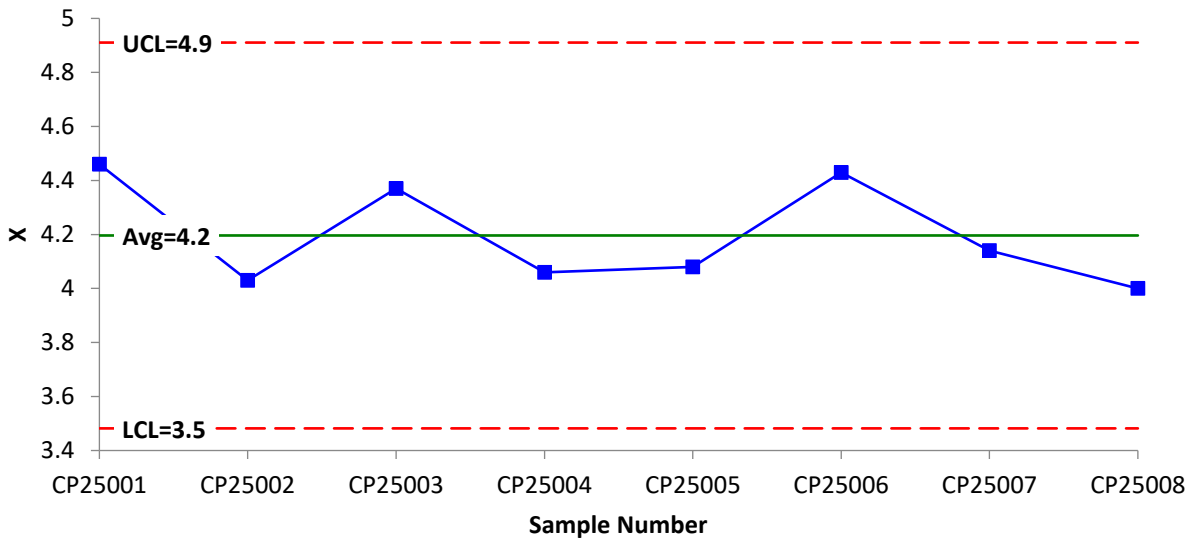
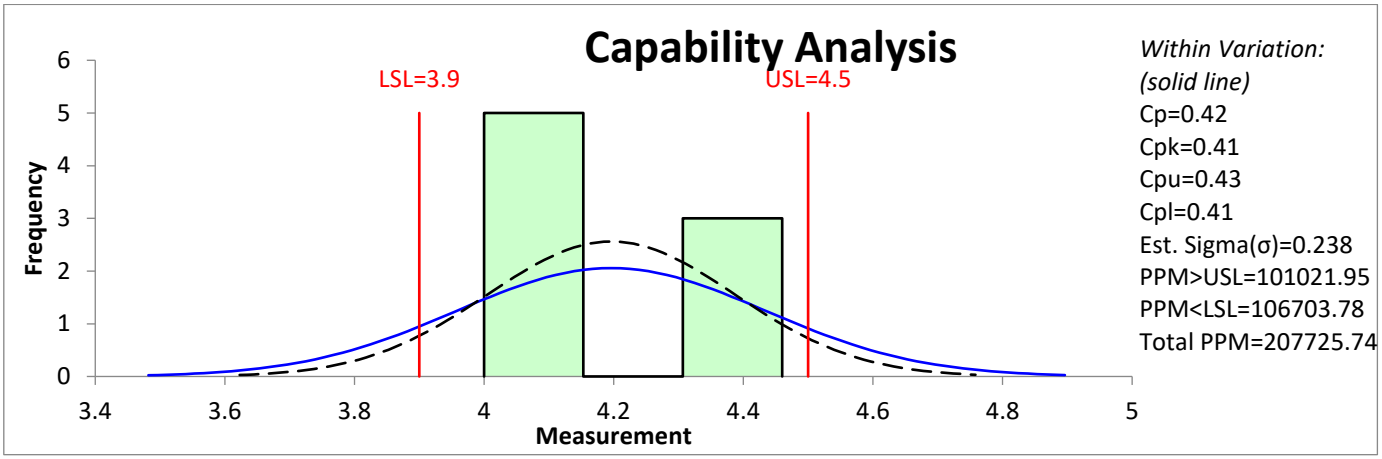
Annexure VII: pH trend of finished product analysis of Ciprofloxacin Injection
Specification of pH (3.90-4.50**)**





Mean	4.2
Standard deviation	0.19115719
Maximum	4.46
Minimum	4
Control limit: 3.5 – 4.9	
Alert limit: 3.82 – 4.58	
Minimum-Maximum: 4 – 4.46	
Alert limit within the specification limit, control limit existing within the specification limit.	

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SUMMARY OF PROCESS CAPABILITY (acceptance criteria: Cpk ≥ 1)

PROCESS CAPABILITY FOR INTERMEDIATE		
1	pH	0.7
2	Rinse Water	4.04
3	Ciprofloxacin	1.8
4	Sodium Chloride	1.92

PROCESS CAPABILITY FOR PRODUCT		
1	pH	0.41
2	Ciprofloxacin	0.82
3	Sodium Chloride	1.19

CQAs	Intermediate		Finished product	
	Control limit	Process capability	Process capability	Control limit
pH	CL outside SL	0.7	0.41	CL outside SL
Ciprofloxacin content	CL within SL	1.8	0.82	CL outside SL
Sodium Chloride content	CL within SL	1.92	1.19	CL within SL

3.4 REVIEW OF BATCHES WITH OUT-OF-SPECIFICATION

No batch was recorded having out of specifications on any quality parameters.

3.5 REVIEW OF CRITICAL DEVIATIONS, NON-CONFORMANCES AND RELATED INVESTIGATIONS

3.5.1 PROCESS DEVIATIONS

No process deviation was reported from the standard processes for batches manufactured during the period Jan., 2025 to Dec., 2025.

3.5.2 YIELD DEVIATIONS

No observed deviation from the predetermined tolerance was observed.

3.5.3 NON CONFORMANCE

No observed non-conformance from the predetermined tolerance was observed.

3.5.4 BATCH FAILURES

There was no batch failure observed during the period Jan., 2025 to Dec., 2025.

3.6 CHANGE CONTROL

There was no change done to the processes, equipment or analytical methods for the manufacture of Ciprofloxacin Intravenous Injection during the year under review. As such, no change control was recorded throughout the period Jan., 2025 to Dec., 2025.

3.6.1 REGULATORY CHANGE

No regulatory changes as well as Master Formula Changes were made concerning these products throughout the manufactured batches till date.

3.6.2 CHANGES IN FORMULATION & PROCESS

No change is made in the formulation as well as process in regular practice. It has been observed that every batch manufactured, following consistent formulation and process, is giving consistent results which are proven by the trend analysis for all the parameters covered and expressed in graphical form.

3.7 STABILITY MONITORING PROGRAMME

The results of the real-time and accelerated stability monitoring program of all the commercial batches carried out showed consistency of all Critical Quality Attributes (CQAs). The product's Critical Quality Attributes (CQAs) like pH, Assay, Appearance, Pyrogenicity and Sterility are all found to be within the set specifications in conformity with the limits of the pharmacopeia. The stability report is attached for reference as BML/QA/SBP/001 (2025)

3.8 QUALITY-RELATED RETURNS, COMPLAINTS AND RECALLS

3.8.1 MARKET COMPLAINTS

Up till now there is no market complaint received for this product during the period Jan. 2025 to Dec. 2025.

3.8.2 RETURNED GOODS FROM MARKET

No finished goods were returned from the market for the period Jan., 2025 to Dec., 2025 due to any quality problem

3.8.3 BATCH RECALLS

Up till now there is no batch recall for product during the period Jan., 2025 to Dec., 2025.

3.8.4 SALVAGED GOODS

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There is no salvaging of goods for the period of Jan., 2025 to Dec., 2025.

3.9 CORRECTIVE ACTIONS

No critical, major or minor deviations have been observed during the year under review. As such, no noteworthy corrective action was necessary. Few observed deviations during the year under review fall under the category of Others, which has to do with colour or print on tertiary packaging.

3.10 VALIDATED ANALYTICAL AND MANUFACTURING PROCEDURES

MANUFACTURING PROCEDURE

Equipment cleaning and sterilization
Dispensing of the raw material
Solution preparation
Batch coding of PVC bag (packaging material)
Solution filling
Sterilization of the filled solution in qualified autoclave
Final packaging of the sterilized solutions

ANALYTICAL PROCEDURE

In-process testing of the cleanliness of equipment
In-process chemical testing of the semi-finished products for critical quality attributes like pH, assay
Chemical testing of the finished products
Microbiological tests on the finished product-
Sterility tests
Bacterial endotoxin tests (Pyrogen and Limulus Amoebocyte Lysate)

SUMMARY FOR PROCESS VALIDATION:

We have manufactured 8 batches of Ciprofloxacin Intravenous Injection 100ml with same manufacturing formula since Jan., 2025 till Dec., 2025 and yearly APQR data shows satisfactory trend throughout the period. Also the retained samples prove that the product manufactured with the existing formulation is stable.

ENVIRONMENTAL CONDITIONS DURING MANUFACTURING OPERATION

The data of environmental conditions during the manufacturing of all batches were reviewed and recorded in the APQR. The environmental conditions of the critical areas were properly monitored in the raw material store, mixing unit, filling unit, batch coding unit and finished goods store

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Environmental Conditions
Std Temp. : NMT 30°C
Relative Humidity: NMT 70%

3.10.1.3 CRITICAL EQUIPMENT PERFORMANCE

The critical equipment used during the manufacturing and analysis of the product were identified and overall performance of these equipment were evaluated by reviewing the breakdown and preventative maintenance records.

3.11 REVIEW OF OBSERVED ADVERSE DRUG REACTIONS

No adverse drug reaction (ADR) was observed with the product during the year under review.

3.11 REVIEW OF PERIODIC RISK-BENEFIT EVALUATION REPORTS

Ciprofloxacin 200mg/100ml Intravenous Injection is an antibacterial preparation formulated for the treatment of infections due to gram-positive and gram-negative bacteria, officially approved for sale in Nigeria since the year 2020. The periodic safety updates reports submitted for the products covering January-December 2025 showed zero risk-benefit ratio (as there was zero risk recorded). All the columns under the spontaneous safety signal and risk report evaluation for the System Organ Class (SOC) are all showing zero. Over 140,000 bags of Bioflex Ciprofloxacin Intravenous Injection was manufactured and distributed across the nation in the year under review and no single report indicating either serious or non-serious early signal detection and/or safety risks on essential system organ class was recorded.

3.13 REVIEW OF POST MARKETING SURVEILLANCE

The reports of clinical and therapeutic performance of the products from major health institutions across the nation have proven the product to be safe and efficacious. No adverse reaction has been reported, no report of product contamination nor pyrogenicity has been reported with the product within the year under review.

3.14 COMPARATIVE PRODUCT QUALITY REVIEW

CONCLUSION

Process capability analysis of critical quality attributes indicates heightened variability levels of control across different stages of the manufacturing process.

- **Sodium chloride content** demonstrates consistent and acceptable process capability at both intermediate and finished product stages, confirming robust control of this parameter.
- **Ciprofloxacin content** shows strong capability at the intermediate stage, indicating adequate control during dispensing and solution preparation. However, reduced capability at the finished

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product stage suggests increased variability during downstream processing steps. This may be influenced by process conditions such as sterilization or product handling, and will require continued monitoring.

- **pH** exhibits low process capability at both intermediate and finished stages, with values below acceptable thresholds, indicating significant process variability. The observed trend suggests that the process is sensitive to factors influencing pH, potentially including raw material variability, solution preparation conditions, or thermal effects during sterilization.

Despite these observations, all batches produced during the review period complied with established specification limits for all tested parameters, and no out-of-specification results were recorded.

The variability observed in pH and finished product ciprofloxacin content will be subject to enhanced trending and monitoring under the ongoing process verification program. Further evaluation may be undertaken to identify opportunities for improved process control, particularly in stages influencing pH stability and assay consistency.

At present, the overall manufacturing process remains capable of producing product meeting predefined quality specifications and is considered acceptable for continued routine production.

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